

Proposed Bid-to-Project Startup Workflow

Web Application Workflow Plan



Workflow structure

One Bid record moves through role-based stages, preserving previously entered information and supporting handoff, review, version history, and final project startup.

STAGE 1**Bid Intake****PURPOSE**

Create the initial bid opportunity using the information available when the bid invitation is first received.

RESPONSIBLE ROLE

Bid Intake / Bid Clerk

Information to Capture**General Project Information**

- Bid / Estimate Number
- Bid Name
- Project Name on Drawings
- Project Address
- City
- State
- Bid Due Date
- Date Invitation Received
- Division / Project Type
 - Insulation
 - Demolition
 - GC
 - Masonry
 - Other
- Bid Type
 - Build-to-Print
 - Design-Build
 - Other
- Budget Bid - Yes / No
- Related Bid / Rebid, if applicable

Project Parties

Capture the information available from the invitation or bid documents:

- Owner
- Architect
- Mechanical Engineer
- General Contractor(s)
- Mechanical Contractor(s)
- Contact Person
- Contact Email
- Contact Phone

The system should allow multiple General Contractors and Mechanical Contractors because the same project may be received from several companies. Multiple bid invitations for the same project should be attached to the same opportunity instead of creating duplicate projects.

Initial Documents

- Invitation Email
- Drawings
- Specifications

- Addenda Received
- Other Bid Documents

Actions

- Save Draft
- Save and Send for Assignment

The first user should only be responsible for entering information reasonably available at the time the invitation is received. Information such as estimator assignments, takeoff personnel, detailed pricing assumptions, and final estimating information should not be required at this stage.

STAGE 2

Bid Assignment

PURPOSE

Determine whether the opportunity will be pursued and assign the estimating team.

RESPONSIBLE ROLES

Estimating Management /
Bid Assignment Role

Information to Capture

- Bid / No Bid Decision
- Priority
- Estimating Team
- Captain / Estimating Lead
- Assistant Estimator
- Bid Clerk
- Internal Estimate Due Date
- Internal Review Due Date

Takeoff Planning

Determine the estimating resources required for:

- Duct
- Hydronic
- Plumbing
- VRF
- Equipment
- Other Scope

The system should allow one or two estimators to be assigned to the same scope when an independent back-check is required.

Actions

- Save Assignment
- Return for Missing Information
- Send to Estimating Setup

STAGE 3**Estimating Setup****PURPOSE**

Complete the technical, wage, project-condition, and pricing information required before takeoff work begins. Much of the information already entered during Bid Intake should appear here automatically.

RESPONSIBLE ROLES

Captain / Estimator /
Assistant Estimator

Project Classification

- Type of Construction
- Construction Subtype
- Company
- MBE / Preference Requirements
- Clearance Requirements
- Government / Private Project
- Federal / State / Local Requirements

Wage and Labor Information

- PLA - Yes / No
- PLA Document
- Wage Decision Number
- Wage Decision Date
- County
- Federal / State / City Wage
- Wage Category
- Base Wage
- Fringe
- Apprenticeship Requirements
- Certified Payroll Requirement
- Citizen / US Person / Real ID Requirements
- Calculated Labor Rate

Where possible, wage information should be selected from maintained lookup tables rather than manually entered for every project.

Project Cost Assumptions

- OCIP / CCIP
- General Liability Coverage
- Workers Compensation Coverage
- Lift Required - Yes / No
- Lift Percentage
- Parking Required
- Parking Reimbursement
- Parking Cost / Percentage
- Sales Tax
- Material Escalation
- Expected Project Start
- Expected Duration
- Expected Project Completion

Insulation Specifications

For Insulation projects, capture the applicable specifications for:

- Hydronic
- Plumbing
- Ductwork
- Piping
- Equipment
- Other Systems

The specification information should be converted into the proposed material/submittal selections that will be used for estimating.

Technical Review

The system should support:

- Prepared By
- Reviewed By
- Review Date
- Approved for Takeoff - Yes / No
- Review Comments

Takeoff work should begin only after the required technical information has been reviewed.

STAGE 4

Takeoff & Estimate Production

PURPOSE

Assign, track, compare, and retain the actual takeoff work.

RESPONSIBLE ROLE

Assigned Takeoff Personnel

Takeoff Categories

Examples include:

- Duct 1
- Duct 2
- Hydronic 1
- Hydronic 2
- Plumbing 1
- Plumbing 2
- VRF
- Equipment
- Other

Each assignment should contain:

- Assigned Person
- Assigned Date
- Due Date
- Status
- Number of Hours Spent
- CSV Upload
- Drawing / PDF Upload

- Takeoff Quantity
- Estimate Number
- Notes

Version Control

Takeoff files should never be overwritten.

Each revision should create a new version:

- Version Number
- Created By
- Created Date
- Reason for Revision
- CSV
- PDF
- Quantity / Result
- Hours Spent

Example: Version 1 > Version 2 > Version 3. The previous versions remain available for historical reference.

Independent Takeoff Comparison

When two estimators are assigned to the same scope, the system should automatically compare their results.

Example:

Duct Takeoff Comparison

- Duct 1 Quantity
- Duct 2 Quantity
- Difference
- Difference %
- Reconciliation Required - Yes / No
- Final Accepted Quantity
- Reviewed By

If the results differ, the responsible team members can review the discrepancy, correct the takeoff, and submit a new version.

STAGE 5

Bid Review & Proposal

PURPOSE

Consolidate the estimate, perform final review, and issue the proposal.

RESPONSIBLE ROLES

Estimating Review /
Proposal Team

Estimate Review

- Final Material Cost
- Final Labor Cost
- Equipment Cost
- Subcontract Cost
- Other Costs

- Total Estimated Cost
- Margin / Markup
- Final Bid Amount
- Scope Included
- Scope Excluded
- Alternates
- Qualifications
- Review Notes

Proposal Versions

Each proposal revision should be retained.

For each version:

- Proposal Version
- Proposal Amount
- Date
- Prepared By
- Reviewed By
- Reason for Change
- Best and Final - Yes / No
- Value Engineering - Yes / No
- Scope Change - Yes / No
- Proposal Attachment

Bid Submission

- Submission Date
- Submission Time
- Submitted Amount
- Submitted By
- Mechanical Contractor
- General Contractor
- Recipient Contact
- Proposal Attachment

Addenda / Amendments

Instead of creating a fixed number of amendment fields, the system should provide:

+ Add Amendment

Each amendment would contain:

- Amendment Number
- Date
- Attachment
- Drawing Changes
- Specification Changes
- Scope Changes
- Wage Impact
- Schedule Impact
- Pricing Impact
- Requires Estimate Revision - Yes / No
- Notes

This allows any number of amendments without creating an unnecessarily large form.

STAGE 6**Post-Bid / Bid Intelligence****PURPOSE**

Track activity after the proposal has been submitted and retain useful market intelligence.

RESPONSIBLE ROLE

Post-Bid / Follow-Up Role

Information to Capture

- Bid Status
- Follow-Up Owner
- Next Follow-Up Date
- Expected Award Date
- Best and Final Requested
- Revised Proposal Required
- General Contractors Still Bidding
- Mechanical Contractors Still Bidding
- Known Insulation Competitors
- Competitor Pricing, if known
- Source of Competitor Information
- Confidence Level of Information
- Customer Feedback
- Current Project Status

Running Project / Bid Log

A running activity log should exist throughout the entire workflow.

Examples:

- Invitation received
- New GC added
- Estimating team assigned
- Addendum received
- Specification changed
- Takeoff submitted
- Takeoff revised
- Proposal submitted
- Best and final requested
- Project postponed
- Project rebid
- Award received
- Lost notification received

Each log entry should capture:

- Date / Time
- User
- Activity Type
- Notes
- Related Attachment, if applicable

This creates a complete history of the opportunity from the first invitation through award or loss.

STAGE 7A**Awarded Project**

If the project is awarded, the existing Bid record should be converted into a Project Startup record rather than creating a new record from scratch.

Award Information

- Job Number
- Date of Award
- Final Contract Amount
- Project Manager
- Mechanical Engineer
- Operations Manager
- Company Performing the Project
- Final Prime Contractor
- Final Mechanical Contractor

Contractual Relationship

The system should support a flexible contractual chain rather than fixed GC/CM fields.

Example:



The application should provide:

+ Add Contractual Layer

For each layer:

- Company
- Relationship / Role
- Tier
- Contact Person
- Project Manager
- Superintendent

- Foreman
- Phone
- Email

This allows the system to accurately represent projects with simple or complex contractual structures.

Project Startup Information

The awarded-project stage would also include the existing startup information such as:

- Form of Contract
- Contract Price
- Labor Budget
- Material Budget
- Equipment Budget
- Bond Cost
- Other / Subcontract Budget
- Total Labor Manhours
- Average Labor Rate
- Projected Start Date
- Projected Completion Date
- Certified Payroll
- OCIP / CCIP
- Bond Information
- Tax Exemption
- Travel / Parking Requirements
- Schedule Received
- Schedule of Values Received
- Special Instructions

Information already collected during bidding should automatically populate these fields and only require confirmation or modification.

STAGE 7B**Lost / No Bid**

If the opportunity is not awarded, the record should remain in the system for historical and market intelligence purposes.

Information to Capture

- Lost / No Bid Date
- Status
 - Lost
 - No Bid
 - Cancelled
 - Postponed
- Awarded Mechanical Contractor
- Awarded Insulation Contractor
- Winning Price, if known
- Our Final Price
- Difference
- Reason Lost
 - Price
 - Scope
 - Schedule
 - Relationship
 - Customer Decision
 - No Bid Decision
 - Other
- Notes
- Possible Future Rebid
- Related Opportunity

Overall User Interface

The system would operate as one Bid record with multiple workflow stages, not independent forms.

A progress bar could appear at the top of the record:

1. Intake > 2. Assignment > 3. Estimating Setup > 4. Takeoff > 5. Proposal > 6. Post-Bid > 7. Award / Lost

Each stage would display:

- Current Owner
- Current Status
- Due Date
- Last Updated By
- Last Updated Date
- Completion %
- Missing Required Items

Primary actions:

Save Draft	Complete My Step & Hand Off	Return to Previous Step
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This approach allows different employees to complete the information they are responsible for without forcing one person to know or enter information that becomes available later.

Items to Confirm Before Finalizing the Workflow

Before completing development, the following items should be confirmed:

1. Which role is officially responsible for Bid Assignment after the initial Bid Intake?
2. Which fields are mandatory before the Bid Intake record can be handed off?
3. Which information may remain incomplete and be filled in by the next role?
4. Who is responsible for preparing the insulation specifications and proposed submittal/material selections?
5. Who performs the second technical review before takeoff begins?
6. Who assigns individual Duct, Hydronic, Plumbing, VRF, and Equipment takeoff personnel?
7. What approval is required before a proposal can be submitted?
8. Who owns the Post-Bid follow-up process?
9. What are the preferred official bid statuses?
10. When a project is awarded, should the Bid automatically convert to a Project, or should management approve the conversion first?
11. Which fields should remain editable after a stage has been completed?
12. Should completed stages be locked, with later corrections handled through revision history?
13. Which notifications should be sent automatically during handoffs and approaching due dates?

Recommended Development Sequence

To reduce complexity and allow review of the workflow before building the entire module, development can proceed in smaller approved sections.

PHASE 1	• Bid Intake • Bid Assignment • Workflow handoff • Basic activity log
PHASE 2	• Estimating Setup • Wage / labor logic • Specifications and technical review
PHASE 3	• Takeoff assignments • File uploads • Version control • Independent takeoff comparison
PHASE 4	• Estimate review • Proposal versions • Addenda • Bid submission
PHASE 5	• Post-Bid Intelligence • Award / Lost workflow
PHASE 6	• Awarded Project Startup • Contractual hierarchy • Final project-team assignments • Additional automation and reporting

This allows each section to be reviewed and refined before the next part is developed while maintaining one consistent workflow from the initial bid invitation through project award and startup.